

$$\text{Loss, } L = \frac{1}{2} (T - Y)^2 \quad (\text{mean square error})$$

$$\text{Let, } u = T - Y$$

$$\therefore L = \frac{1}{2} u^2$$

using chain rule,

$$\frac{\partial L}{\partial Y} = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial Y}$$

First,

$$\frac{\partial L}{\partial u} = \frac{\partial}{\partial u} \left(\frac{1}{2} u^2 \right) = u$$

second,

$$\frac{\partial u}{\partial Y} = \frac{\partial}{\partial Y} (T - Y) = -1$$

$$\therefore \frac{\partial L}{\partial Y} = u(-1) = -(T - Y)$$

$$\Rightarrow \frac{\partial L}{\partial Y} = Y - T$$

$$Y = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1}$$

$$\frac{\partial Y}{\partial z} = -1(1+e^{-z})^{-2} \cdot \frac{\partial}{\partial z}(1+e^{-z})$$

$$= -1(1+e^{-z})^{-2} \cdot (-e)^{-z}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2} \cdot \frac{1+e^{-z}}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-z}} \cdot \left(1 - \frac{1}{1+e^{-z}}\right)$$

$$\therefore \frac{\partial Y}{\partial z} = Y(1-Y)$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial Y} \cdot \frac{\partial Y}{\partial z}$$

$$= (Y-\tau) \cdot Y(1-Y)$$